FUTURA

Calculator Engine

REFERENCE DOCUMENT

A complete guide to the retirement planning engine — its methodology, mathematical models, scoring framework, and output interpretation.

Document Purpose

This reference document describes the complete specification of the Futura Calculator Engine in plain language. It explains every concept, formula, and decision the engine makes — from validation through to final scoring. No prior technical knowledge is required to read this document.

Audience: product managers, financial analysts, QA engineers, and implementation partners integrating the Futura platform.

1. Purpose of the Calculator Engine

The Calculator Engine is the financial brain of the Futura platform. Its job is to answer one question with precision: given everything we know about a user — their age, income, savings, and spending habits — will they have enough money to retire comfortably?

The engine does not give a simple yes or no. Instead, it produces a rich picture of retirement readiness: a concrete savings target, a probability of success under thousands of simulated futures, and a health score that identifies exactly which part of a person's financial behaviour is putting their retirement at risk.

The engine is built around a specific concept called Retirement Blindness — the state in which a person's retirement plan is fundamentally broken without them realising it. Poor assumptions, insufficient savings, or an unrealistic timeline can each cause blindness. The engine's primary design goal is to surface this condition early, clearly, and without ambiguity.

What is Retirement Blindness?

Retirement Blindness occurs when a person believes their savings plan is adequate, but the underlying mathematics of their situation — accounting for inflation, market volatility, and life expectancy — shows otherwise. A person can be earning well, saving regularly, and still be blind

if their assumptions about returns are too optimistic or their expenses too high relative to their savings rate.

The engine is designed to make this invisible problem visible.

2. Inputs: What the Engine Needs to Know

The engine requires a complete picture of the user's financial situation and planning assumptions. All inputs are mandatory. The engine will not produce a result if any input is missing or violates its validation rules.

2.1 Personal Timeline

These three values define the time structure of the retirement plan and form the backbone of every calculation.

Field	What It Represents	Why It Matters
Current Age	The user's age today.	Determines the accumulation period — the number of years the user has left to save and invest before retirement.
Retirement Age	The age at which the user intends to stop working.	Sets the end of the savings phase and the beginning of the withdrawal phase. A later retirement age dramatically improves outcomes by extending accumulation and shortening withdrawal.
Life Expectancy	The age the user plans the financial model around.	Defines how long the retirement corpus must last. The engine plans to the full life expectancy — it does not assume an average. This is the safest approach to avoid outliving savings.

2.2 Income and Spending

The engine needs to understand the user's current financial flows — how much they earn, how much they spend, and how much they put aside every month.

Field	What It Represents	Why It Matters
Monthly Income	Total gross monthly income from all sources.	Serves as the baseline for calculating discipline ratios. A higher income with a low investment rate is just as risky as a low income.
Monthly Expenses	Total monthly outgoings including living costs, EMIs, and discretionary spending.	The engine inflates this figure to retirement age to establish the lifestyle cost the corpus must support. This is the single most important driver of the corpus target.

Field	What It Represents	Why It Matters
Current Savings	The total investment corpus the user has already accumulated — across all instruments.	Provides a head-start on the corpus target. The engine compounds this figure over the accumulation period at the assumed return rate.
Monthly Investment	The amount the user invests every month as a Systematic Investment Plan (SIP).	Determines how much the corpus grows through regular contributions. Even modest monthly investments, sustained over decades, create compounding that rivals lump-sum savings.

2.3 Return and Inflation Assumptions

These are the economic assumptions that the engine uses. They come in two forms: deterministic rates for the base-case projection, and ranges for the Monte Carlo simulation that tests plan robustness.

Field	What It Represents	Typical Range
Return Rate (deterministic)	The fixed annual rate of return assumed for all investments in the base-case projection.	6% – 12% depending on asset mix
Inflation Rate (deterministic)	The fixed annual rate at which living costs are expected to rise.	4% – 7% for most economies
Return Min / Return Max	The lower and upper bounds of the return rate range used in Monte Carlo simulation. Captures best and worst market environments.	e.g. 4% – 15%
Inflation Min / Inflation Max	The lower and upper bounds of the inflation rate range used in Monte Carlo simulation. Captures low and high inflation environments.	e.g. 2% – 8%

Important: The deterministic rates determine the engine's single best-guess projection. The Monte Carlo ranges determine how 'stressed' the plan is tested. Wider ranges produce a more conservative probability estimate and give a better picture of true risk.

2.4 Simulation Count

The engine runs the retirement scenario thousands of times using different combinations of returns and inflation, each time checking whether the outcome is a success or failure. The number of simulations controls how accurate this probability estimate is.

> **Minimum Simulations: 1,000**
>
> The engine enforces a minimum of 1,000 simulation runs. Below this threshold, the probability estimate becomes statistically unreliable — a run of just 100 simulations could easily over- or

under-estimate success probability by 10 or more percentage points.

The recommended default is 2,000 simulations, which balances accuracy with performance. For critical assessments, 5,000 or more may be used.

3. Validation: Rules the Engine Enforces

Before performing any calculation, the engine checks that every input makes financial and logical sense. If any rule is violated, the engine stops immediately and reports the specific problem. This is by design — running calculations on an impossible scenario would produce meaningless results.

Rule	The Condition Required	What Happens If Violated
Age ordering	Retirement age must be strictly greater than current age.	An engine cannot project a savings period of zero or negative years. The plan has no accumulation phase.
Longevity ordering	Life expectancy must be strictly greater than retirement age.	If life expectancy is not beyond retirement, there is no withdrawal phase to plan for. The model collapses.
Budget discipline	Monthly expenses must be less than monthly income.	If expenses equal or exceed income, there is no capacity to save. The corpus will never grow; the plan is impossible.
Simulation floor	At least 1,000 simulations must be requested.	Fewer simulations produce statistically unreliable probability estimates. The engine refuses to report inaccurate odds.
Return range integrity	The minimum return rate must be less than the maximum.	An inverted range is logically invalid and would cause the random sampling process to fail.
Inflation range integrity	The minimum inflation rate must be less than the maximum.	Same as above — an inverted inflation range cannot be sampled from meaningfully.

4. How the Engine Calculates: The Four Phases

Once inputs are validated, the engine runs four sequential phases. Each phase builds on the previous one. The first two establish the ‘base case’ using fixed assumptions. The third stress-tests that base case with thousands of alternative futures. The fourth converts all of this into the Futura scores.

1 **Input Validation**

Enforces financial boundary conditions. No calculation proceeds until all rules pass.

2 **Deterministic Projection**
Calculates the Magic Number corpus target and projected savings using fixed rates.

3 **Monte Carlo Simulation**
Runs thousands of alternative futures using variable rates to compute a probability of success.

4 **Scoring and Classification**
Converts raw numbers into the Blindness Score and Status classification.

4.1 Phase Two: Deterministic Projection

The deterministic projection is the engine's base-case calculation. It uses the single fixed return rate and inflation rate provided as inputs, and calculates three core outputs: the cost of retirement, the money needed to fund it, and the money the user is on track to have.

Step A — Inflation-Adjusted Future Monthly Expense

A person's current monthly expenses are not the right number to plan retirement around. Because prices rise every year due to inflation, the same lifestyle will cost significantly more in 20 or 30 years. The engine calculates exactly how much more.

The current monthly expense is compounded forward by the inflation rate for every year between now and retirement. If someone currently spends ₹30,000 per month and inflation averages 5%, their lifestyle will cost approximately ₹77,000 per month in 20 years.

FUTURE MONTHLY EXPENSE

Future Expense = Current Expense × (1 + Inflation Rate) ^ Years to Retirement

This is the monthly cost of living that the retirement corpus must be able to sustain from day one of retirement.

Step B — The Corpus Needed (The Magic Number)

The corpus needed is the total lump sum that must exist in the user's portfolio on the day they retire. This is the single most important number the engine produces — referred to internally as the Magic Number.

To calculate it, the engine uses the concept of the Real Rate of Return — the return that remains after inflation is stripped away. This is important because both the corpus and the expenses it must fund grow together over retirement. Using the nominal return alone would overstate the corpus's purchasing power.

REAL RATE OF RETURN

Real Rate = [(1 + Return Rate) ÷ (1 + Inflation Rate)] − 1

This is the return in excess of inflation — the rate at which wealth grows in real, inflation-adjusted terms.

Using the real rate, the engine calculates the corpus as the present value of an annuity — a stream of monthly payments for the entire length of retirement. In plain terms: how large a pot of money, growing at the real rate, is required to sustain the future monthly expense for every month from retirement until life expectancy?

CORPUS NEEDED (MAGIC NUMBER)

Corpus Needed = Future Monthly Expense × 12 × Annuity Factor (Real Rate, Retirement Years)

The annuity factor accounts for both the duration of retirement and the compounding growth of the remaining corpus during withdrawals.

i

Why the Real Rate Matters

Using the nominal return rate to calculate the required corpus understates the corpus needed, because it ignores the fact that expenses will keep rising throughout retirement. The real rate — return minus inflation — is the only honest measure of whether a corpus is growing fast enough to keep up with the cost of living.

A portfolio earning 10% returns in an 8% inflation environment is effectively only growing at 1.85% in real terms — a very different picture.

Step C — Projected Future Savings

The engine also calculates how large the user's investment portfolio is projected to be on the day they retire, assuming the deterministic return rate holds throughout the accumulation period. This is compared directly against the Magic Number to assess the gap.

The future savings figure has two components that are added together:

- **Lump-sum growth:** The user's current savings compounded at the monthly return rate for every month until retirement. Money already saved has the longest time to grow.
- **SIP growth:** The future value of the monthly investment stream. Every monthly contribution is compounded from the month it is made until retirement. This is the engine of long-term wealth building.

PROJECTED FUTURE SAVINGS

Future Savings = [Current Savings × (1 + r)^n] + [Monthly Investment × ((1 + r)^n − 1) ÷ r]

Where r = monthly return rate (annual rate ÷ 12) and n = total months to retirement.

✓ **Interpreting the Gap**

If Future Savings exceeds the Corpus Needed, the plan has a surplus — the user is on track. If Future Savings falls short, the plan has a deficit. The deficit is the gap the user must close by saving more, reducing expenses, extending their working years, or accepting a more modest lifestyle in retirement.

The gap is shown explicitly in the Futura output alongside the component figures.

4.2 Phase Three: Monte Carlo Simulation

The deterministic projection answers: 'What happens if everything goes exactly as assumed?' The Monte Carlo simulation answers the more important question: 'What is the probability of success across the full range of things that could realistically happen?'

Markets do not deliver a smooth, predictable return every year. Inflation fluctuates. The sequence in which good and bad years occur matters enormously. Monte Carlo simulation models all of this by running the retirement calculation thousands of times, each time with a different set of randomly sampled economic conditions.

How Random Rates Are Generated: The Box-Muller Transform

In each simulation run, the engine does not use the fixed rates. Instead, it samples new return and inflation rates from within the user-specified minimum and maximum ranges. The sampling uses a mathematical technique called the Box-Muller Transform.

The Box-Muller Transform converts two uniformly distributed random numbers into a pair of numbers that follow a normal (bell-curve) distribution. This is important because real-world investment returns are not uniformly random — they cluster around an average, with extreme outcomes (very high or very low returns) being less frequent than moderate ones.

≈ **Why Normal Distribution?**

A normal distribution means that most simulated return years cluster around the middle of the [min, max] range, with genuinely extreme years appearing less often — which is exactly how real markets behave over time. Using a simple uniform distribution would overweight extreme outcomes and produce an artificially pessimistic probability estimate.

The transform is scaled so that virtually all sampled values fall within the user-specified [min, max] range, respecting the intended bounds while still producing realistic variance.

How Success is Counted

Each simulation run uses a freshly sampled return rate and inflation rate to re-run the full deterministic projection from scratch. At the end of the run, the engine asks one question: does the projected future savings meet or exceed the corpus needed?

- If yes: the run is counted as a success.
- If no: the run is counted as a failure.

After all simulations are complete, the engine divides the number of successful runs by the total number of runs to produce the Probability of Success, expressed as a percentage.

PROBABILITY OF SUCCESS

Probability = (Number of Successful Runs ÷ Total Simulation Runs) × 100

A probability of 85% means the plan succeeded in 850 out of 1,000 simulated futures.

A high probability means the plan holds up even when markets underperform or inflation spikes. A low probability means the plan is fragile — it only works in the most favourable scenarios and will likely fail under realistic market stress.

5. The Futura Scoring System

The engine produces two proprietary Futura metrics that translate the raw financial calculations into actionable signals. These metrics are designed to be complementary — one is quantitative (the probability of success from Monte Carlo), the other is a qualitative behavioural health score (the Blindness Score).

5.1 Probability of Success

The probability of success is the most direct measure of whether a retirement plan is likely to work. It represents the percentage of simulated futures in which the user's projected savings were sufficient to fund the required corpus.

Unlike the deterministic projection, which gives a single answer based on assumed conditions, the probability accounts for the full range of economic environments that could realistically unfold. A plan with a 90% probability of success is not just on track — it is robust enough to withstand a wide range of bad scenarios.

Probability Band	What It Means in Practice
≥ 80%	The plan is strong. Even in a broad range of unfavourable market conditions, the corpus target is reached. This is the target zone.
60% – 79%	The plan is workable but exposed to meaningful downside. A prolonged bear market or sustained inflation above the assumed range could cause it to fall short.
40% – 59%	The plan is fragile. It succeeds only in relatively favourable economic conditions. Failure is the more likely outcome under realistic stress.
Below 40%	The plan is statistically likely to fail. It is dependent on an unusually favourable sequence of events and should be treated as fundamentally unsound.

5.2 The Retirement Blindness Score

The Blindness Score is a number between 0.0 and 1.0 that measures the overall health of a retirement plan from a behavioural perspective. Where the probability of success asks 'Will this plan work?', the Blindness Score asks 'Is this person's financial behaviour compatible with a successful retirement?'

A score of 1.0 indicates a perfectly structured retirement plan. A score approaching 0.0 indicates that the user is in the Retirement Blindness condition — their plan is structurally unsustainable regardless of market conditions. The score is designed to be shown as a visual health indicator in the Futura UI.

How the Score is Constructed

The Blindness Score is a weighted average of four separate components. Each component measures a different dimension of financial health and is scored between 0.0 and 1.0. The four components and their weights are:

Weight	Component	What It Measures and Why It Matters
40%	**Savings-to-Corpus Ratio**	Measures how close projected future savings are to the required corpus. This carries the highest weight because it directly reflects whether the user is on track to hit the Magic Number. A ratio of 1.0 means savings exactly meet the corpus; above 1.0 is a surplus; below is a deficit.
20%	**Investment-to-Income Ratio**	Measures how much of monthly income is being invested. A higher ratio reflects stronger savings discipline. Even a high income contributes little if only a small fraction is invested. This component rewards consistent investment behaviour.
20%	**Expense-to-Income Discipline**	Measures how well the user lives within their means. It is calculated as one minus the expense ratio. A user spending 90% of income scores 0.10 on this component; a user spending 50% scores 0.50. High expenses leave no room for savings and compress the corpus.
20%	**Time Horizon — Age Factor**	Measures how many accumulation years remain. Calculated relative to a 40-year reference horizon. A user starting at 25 has maximum time advantage; one starting at 55 has very little. Time is the most powerful variable in compounding, and this component captures its presence or absence.

BLINDNESS SCORE (COMPOSITE)

Score = (0.40 × Savings Ratio) + (0.20 × Investment Ratio) + (0.20 × Expense Discipline) + (0.20 × Time Horizon)

Each component is clamped between 0.0 and 1.0 before weighting. The final score is therefore always in the range 0.0 to 1.0.

i

Why 40% Weight on Savings-to-Corpus?

The savings-to-corpus ratio is the most outcome-proximate measure — it directly measures the gap between where the user is headed and where they need to be. The other three components

are leading indicators: they capture the behaviours that drive the savings ratio over time. A user could theoretically score well on the three behavioural components yet still have a low savings ratio if they started very late; the 40% weight ensures the score reflects actual trajectory, not just good habits.

6. Engine Outputs and Status Classification

The engine produces a complete RetirementOutputs object. The table below describes each output field, what it represents, and how it should be interpreted.

Output Field	What It Is	How to Interpret It
Corpus Needed	The Magic Number — the total retirement corpus required to sustain the inflation-adjusted lifestyle for the full retirement period.	This is the user's savings target. All other outputs are meaningful only in relation to this number.
Future Savings	The projected total portfolio value at retirement, combining compounded existing savings and the SIP stream.	Compare directly to Corpus Needed. A surplus means the plan is on track; a deficit shows the shortfall the user must address.
Future Expense	The inflation-adjusted monthly cost of living at the start of retirement.	This is the 'lifestyle price tag' in retirement-year money. It helps users understand the real cost of their current spending habits.
Probability	The percentage of simulated futures in which the plan succeeded.	The primary risk metric. Interpret as: 'Under realistic market volatility, this plan succeeds X% of the time.'
Blindness Score	The composite behavioural health score (0.0 – 1.0).	A qualitative signal for the UI. Use to identify which of the four components is dragging the score down and guide the user toward the right corrective action.
Status	The plain-language classification of the plan.	The top-level label. Derived from the combination of Probability and Blindness Score. Four values: Safe, Moderate, High Risk, Blindness.

6.1 Status Classification

The Status field provides the plain-language summary of the plan's health. It is derived from both the Probability of Success and the Blindness Score together. A plan must perform well on both dimensions to achieve Safe status.

Status	Probability	Blindness Score	Recommended Action

Status	Probability	Blindness Score	Recommended Action
SAFE	≥ 80%	0.75 – 1.00	Maintain current strategy. Annual review recommended.
MODERATE	60% – 79%	0.50 – 0.74	Review asset allocation. Consider incrementally increasing the monthly investment.
HIGH RISK	40% – 59%	0.25 – 0.49	Significantly increase SIP contributions or extend the retirement horizon by 3–5 years.
BLINDNESS	< 40%	0.00 – 0.24	Immediate intervention required. Reassess all core assumptions with a financial advisor.

⚠ **Blindness vs. High Risk**

High Risk and Blindness may appear similar but represent fundamentally different situations. High Risk means the plan is structurally possible but requires significant corrective action. Blindness means the core assumptions are incompatible with a sustainable retirement — the plan is not salvageable through incremental adjustment alone. A user in Blindness typically needs to revisit their retirement age, expected lifestyle, or both.

7. Worked Example: Interpreting a Real Output

The following example walks through a complete engine run using a realistic profile, showing how each input drives each output.

7.1 Example Profile

Input	Value	Notes
Current Age	30	30 years of accumulation ahead
Retirement Age	60	30-year savings horizon
Life Expectancy	85	25-year retirement phase
Monthly Income	5,000	All figures in consistent currency units
Monthly Expenses	2,000	40% expense ratio — disciplined
Current Savings	10,000	Modest head-start corpus
Monthly Investment	1,000	20% of income invested
Inflation Rate (deterministic)	5%	Moderate inflation assumption
Return Rate (deterministic)	10%	Equity-oriented portfolio assumption
Return Range (Monte	4% – 15%	Wide range to stress-test robustness

Input	Value	Notes
Carlo)		
Inflation Range (Monte Carlo)	2% – 8%	Captures low and high inflation environments
Simulations	2,000	Well above the 1,000 minimum

7.2 Engine Reasoning, Step by Step

Step 1: Future Monthly Expense

Current expenses of 2,000 compounded at 5% inflation for 30 years becomes approximately 8,644 per month. This is the lifestyle cost the corpus must sustain from the first day of retirement.

Step 2: Corpus Needed

At a real rate of return of approximately 4.76% (10% return minus 5% inflation), the engine calculates the present value of 8,644 per month for 25 years of retirement. The resulting Magic Number is approximately 1,620,000.

This means the user must have roughly 1.62 million in their portfolio the day they retire to fund their retirement without running out of money, even accounting for ongoing inflation.

Step 3: Future Savings

The current savings of 10,000 compounded at 10% annually for 30 years grows to approximately 174,500. The monthly SIP of 1,000 invested for 30 years at 10% grows to approximately 2,260,000. Combined, future savings are projected at around 2,434,500 — comfortably above the corpus target of 1,620,000.

Step 4: Monte Carlo Simulation

Across 2,000 simulated futures with returns ranging from 4% to 15% and inflation from 2% to 8%, the plan succeeds in approximately 1,760 out of 2,000 runs. Probability of success: 88%.

Step 5: Blindness Score

The four components score as follows:

- **Savings-to-Corpus Ratio (40% weight):** 2,434,500 ÷ 1,620,000 = 1.50, clamped to 1.0. Weighted contribution: 0.40.
- **Investment-to-Income Ratio (20% weight):** 1,000 ÷ 5,000 = 0.20. Weighted contribution: 0.04.
- **Expense Discipline (20% weight):** 1 − (2,000 ÷ 5,000) = 0.60. Weighted contribution: 0.12.
- **Time Horizon (20% weight):** 30 years ÷ 40 = 0.75. Weighted contribution: 0.15.

BLINDNESS SCORE FOR THIS PROFILE

0.40 + 0.04 + 0.12 + 0.15 = 0.71

Score of 0.71 falls in the Moderate band (0.50 – 0.74) despite an 88% probability of success.

Step 6: Status

Probability of 88% maps to Safe. Blindness Score of 0.71 maps to Moderate. The engine applies the more conservative of the two — or takes both into account — to produce a final status of Moderate. The user is likely to succeed, but the relatively low investment-to-income ratio (only 20% of income invested) is flagging a vulnerability that could become material if circumstances change.

✓ **Key Insight from This Example**

This profile is healthy overall, but the Blindness Score is held back primarily by the investment-to-income ratio (0.20 component score). Increasing the monthly SIP from 1,000 to 1,500 would raise this component score from 0.20 to 0.30 and lift the overall Blindness Score above 0.75, moving the status to Safe. The engine's component breakdown makes it clear exactly what to fix.

8. Design Principles and Known Limitations

8.1 Why Two Analytical Methods?

The deterministic projection and the Monte Carlo simulation serve different purposes and are deliberately both included.

The deterministic projection gives users a concrete, understandable target: 'You need X to retire.' It is simple, auditable, and motivating. The Magic Number is a specific figure users can act on.

The Monte Carlo simulation gives a probabilistic confidence estimate that the deterministic method cannot provide. Two users with identical Magic Numbers and identical projected savings could have very different risk profiles depending on the volatility of their assumed return range. The probability of success captures this difference.

Together, the two methods eliminate the two most common failure modes of retirement calculators: false precision (a single scenario that looks exact but ignores variance) and false uncertainty (probability bands so wide they are not actionable).

8.2 Known Limitations

The engine is designed for clarity and reliability within its stated scope. The following limitations apply:

- **Sequence-of-returns risk is approximated, not explicitly modelled.** The Monte Carlo simulation samples average rates per run but does not model the specific damage caused by a severe bear market in the early years of retirement, which is the most dangerous form of volatility for a retiree.
- **Tax treatment is not modelled.** All figures are pre-tax. The engine does not differentiate between tax-advantaged and taxable accounts, nor does it model withdrawal tax liability in retirement.
- **Currency is agnostic.** The engine operates in whatever currency the inputs are denominated in. No currency conversion, purchasing power parity, or country-specific tax rules are applied.
- **Healthcare inflation is not separated.** Medical costs typically inflate faster than general living expenses, particularly in later retirement. The engine uses a single inflation rate for all expenses.
- **Withdrawal strategy is fixed as an annuity model.** The engine assumes the corpus is drawn down as a level annuity adjusted for inflation. Dynamic withdrawal strategies (such as Guyton-Klinger rules or variable percentage withdrawal) are not modelled.

8.3 Planned Future Enhancements

The following enhancements are on the product roadmap and will extend the engine's analytical capability:

- Tax-aware projections distinguishing between tax-deferred, tax-exempt, and taxable investment accounts.
- Dynamic withdrawal rate modelling based on Safe Withdrawal Rate research and adaptive spending rules.
- Healthcare cost separation with a configurable healthcare inflation rate for users above 60.
- Sequence-of-returns risk modelling with explicit simulation of early-retirement bear market scenarios.
- Goal-based scenario comparison allowing users to model multiple retirement ages or lifestyle assumptions side by side.

9. Glossary of Key Terms

Term	Definition
Accumulation Period	The years between today and retirement during which the user saves and invests. The longer this period, the more compounding can work.
Annuity	A series of equal periodic payments. The engine models retirement withdrawals as a monthly annuity funded by the corpus.
Box-Muller Transform	A mathematical technique used to generate normally distributed random numbers from uniformly distributed inputs. Used by the engine to sample realistic return and inflation values in each Monte Carlo run.
Corpus	The total investment portfolio value. In this document, used to refer to the retirement savings pool.
Corpus Needed (Magic Number)	The total lump-sum amount that must exist in the portfolio on the day of retirement to sustain the desired lifestyle until life expectancy.
Deterministic Projection	A calculation using fixed, assumed rates — producing a single, concrete

Term	Definition
	outcome rather than a range.
Magic Number	See Corpus Needed. The term used in the Futura UI to make the concept accessible to users.
Monte Carlo Simulation	A method of modelling uncertainty by running thousands of iterations with randomly sampled variables to produce a probability distribution of outcomes.
Normal Distribution	A bell-shaped probability distribution where outcomes cluster around a central average with fewer occurrences at the extremes. Used by the engine to model realistic return variance.
Real Rate of Return	The investment return after adjusting for inflation. Represents the true growth in purchasing power.
Retirement Blindness	The engine's core detection target: the condition in which a person's retirement plan appears adequate on the surface but is fundamentally unsustainable under realistic financial conditions.
SIP (Systematic Investment Plan)	A regular, periodic investment contribution — equivalent to a monthly investment into a mutual fund or portfolio.
Status	The plain-language classification of a retirement plan produced by the engine: Safe, Moderate, High Risk, or Blindness.
Withdrawal Phase	The years from retirement until life expectancy during which the corpus is drawn down to fund living expenses.

Futura Platform • Calculator Engine Reference Document • v1.0

This document reflects the canonical engine specification. It is maintained alongside the module and updated on each version release.